Ajib Paudel

Data Scientist | Machine Learning & Statistical Modeling | Python, PyTorch, SQL

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github.com/paudelajib | linkedin.com/in/paudelajib | Google Scholar

SUMMARY

Physics PhD with 10+ years of hands-on data science: analyzing large datasets, building statistical models, and quantifying uncertainty. Recently builds deep learning models in PyTorch, including graph neural networks that beat the prior benchmark by 5% on a hard classification task. Strong in feature engineering, error analysis, and benchmarking.

TECHNICAL SKILLS

Machine learning: PyTorch, scikit-learn, XGBoost, graph neural networks, classification, regression, feature engineering, model evaluation (ROC-AUC, lift, confusion matrices), GPU training

Statistics: Statistical inference, Monte Carlo simulation, uncertainty quantification, calibration and bias correction, time series and walk-forward backtesting

Programming & data: Python (pandas, NumPy), SQL, C++, Bash, Jupyter, HDF5

Computing & tools: Linux, Git, HPC batch workflows, GPU clusters, LLM APIs (Claude), AI-assisted development

EXPERIENCE

Research Associate | Fermi National Accelerator Laboratory (Fermilab)

Batavia, IL | May 2021 - September 2026

- Trained graph neural networks (PyTorch) for multi-class classification and regression on particle-detector sensor data, improving separation of two look-alike event classes by **5%** over the prior benchmark from another group.
- Engineered features, validated labels, and designed train/validation/test splits for simulated datasets of up to 40,000 events, processed through staged HDF5 pipelines on shared HPC/GPU clusters.
- Extended the shared codebase and benchmarked model variants, input features, and loss functions using confusion matrices, class probabilities, and event-level error analysis.
- Led a joint working group of two detector teams developing a neutron-capture calibration method; presented to working groups of a **1,500+ member** international collaboration; mentored students.

Graduate Research Assistant / Teaching Assistant | Kansas State University

Manhattan, KS | August 2015 - May 2021

- Measured particle interaction rates (pion-argon cross sections) from CERN ProtoDUNE-SP beam data: event selection, simulation-vs-data comparison, statistical estimation, and systematic uncertainties.
- Led detector energy calibration using cosmic-ray muons, building corrections for spatial non-uniformity in detector response; wrote analysis code in Python, C++, and ROOT; taught undergraduate physics.

SELECTED PROJECTS

Credit Default Risk Pipeline | Python, pandas, SQLite, scikit-learn, XGBoost | github.com/paudelajib/credit-risk-pipeline

- Built a reproducible pipeline for 30,000 credit-card records: data-quality checks, SQL analysis, preprocessing, and model comparison in a Jupyter notebook.
- Compared logistic regression and XGBoost with stratified splits and early stopping; XGBoost reached **0.778 ROC-AUC** on 6,000 held-out customers and caught 30.3% of defaults in the riskiest 10% (**3.03x lift**).

S&P 500 Signal Scanner | Python, pandas, scikit-learn | github.com/paudelajib/call-signal-scanner2 | **Live demo**

- Ranks all S&P 500 stocks with 1-week, 1-month, and 6-month buy scores from 17 daily features (momentum, trend, sector-relative P/E, rate and inflation sensitivity, news sentiment); public demo auto-refreshes each weekday.
- Validated with a walk-forward time-series backtest since 2016 (information coefficient, quintile spreads, out-of-sample check) and a no-lookahead test; next step is training ML models against this baseline.

EDUCATION

Ph.D. in Physics, Kansas State University | May 2021

M.Sc. in Physics, Tribhuvan University, Nepal | 2012 · **B.Sc. in Physics**, St. Edmund's College, India | 2008

Publications: 60+ co-authored peer-reviewed papers, including *Physical Review D* and *JINST* (Google Scholar)

Training: Databases and SQL for Data Science with Python (IBM, 2026); GenAI for Algorithmic Trading (2025)